MATHEMATICS

SECTION – I
Single Correct Choice Type

This section contains 8 single correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

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1. Variable plane $lx + my + nz = p$ (normal form with p constant) intersects the co-ordinate axes at A, B, and C respectively, then the least area of ΔABC, is

(a) $\frac{27}{4}p^2$ (b) $\frac{3\sqrt{3}}{2}p^2$

(c) $\frac{p^2}{3\sqrt{3}}$ (d) none of these

2. Two regular polygons are inscribed in same circle. The first polygon has 1982 sides and second has 2973 sides. If the polygons have any common vertices, the number of such vertices are

(a) 1 (b) 990 (c) 991 (d) none of these

3. The curve represented by the equation $\frac{x^2}{\sin\sqrt{2} - \cos\sqrt{3}} + \frac{y^2}{\sin\sqrt{3} - \cos\sqrt{2}} = 1$ is

(a) an ellipse with foci on x-axis
(b) an ellipse with foci on y-axis
(c) an hyperbola with foci on x-axis
(d) an hyperbola with foci on y-axis

4. The value of $\sum_{r=1}^{n} \frac{r}{4r^4 + 1}$ is equal to

(a) $\frac{\sum n}{4\sum n + 3}$ (b) $\frac{\sum n}{2\sum n + 3}$ (c) $\frac{\sum n}{8\sum n + 3}$ (d) $\frac{\sum n}{4\sum n + 1}$

5. The least value of k such that the equation $(\ln x) + k = e^{x-k}$ has a solution is

(a) e (b) $1/e$ (c) 1 (d) none of these

6. Let A, B and C be square matrices of order 3×3 with real elements. If A is invertible and $(A - B)C = BA^{-1}$, then

(a) $C(A-B)=A^{-1}B$ (b) $C(A-B)=BA^{-1}$

(c) $(A-B)C=A^{-1}B$ (d) all the above

7. Given two real sets $A=\{a_1, a_2, ...a_{2n}\}$ and $B\{b_1, b_2, ...b_n\}$. If $f:A\to B$ is a function such that every element of B has pre-image and $f(a_1)\le f(a_2)\le .....\le f(a_{2n})$, then number of such mappings are

(a) ${}^{2n}C_n$ (b) ${}^{2n}C_{n-1}$

(c) ${}^{2n-1}C_{n-1}$ (d) none of these

8. If $x_1=\sqrt{3}$ and $x_{n+1}=\dfrac{x_n}{1+\sqrt{1+x_n^2}}$, $\forall n\in N$, then $\lim\limits_{n\to\infty}2^n x_n$ equals to

(a) $\dfrac{3}{2\pi}$ (b) $\dfrac{2}{3\pi}$

(c) $\dfrac{2\pi}{3}$ (d) $\dfrac{3\pi}{2}$

SECTION – II
Multiple Correct Choice Type

This section contains 4 multiple correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONE OR MORE** is/ are correct.

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9. Let $S_n=\lim\limits_{n\to\infty}\left(\dfrac{1}{n^6}+\dfrac{32}{n^6}+.....+\dfrac{1}{n}\right)$ and $T_n=\lim\limits_{n\to\infty}\left(\dfrac{1}{n^6}+\dfrac{32}{n^6}+.....+\dfrac{(n-1)^5}{n^6}\right)$, then which of the following is/ are true?

(a) $S_n\to\dfrac{1}{6}^+$ (b) $(S_n+T_n)<\dfrac{1}{3}$

(c) $(S_n+T_n)>\dfrac{1}{3}$ (d) $T_n\to\dfrac{1}{6}^-$

10. $z_1, z_2.....z_n$ be non-zero complex numbers of equal modulus and satisfying the equation $z^n-|z|z^{n-1}+1=0$, then

(a) $\text{Re}\left(\sum_{j=1}^{n}\sum_{k=1}^{n}\frac{z_j}{z_k}\right)=0$ (b) $\text{Re}\left(\sum_{j=1}^{n}\sum_{k=1}^{n}\frac{z_j}{z_k}\right)=1$

(c) $\text{Im}\left(\sum_{j=1}^{n}\sum_{k=1}^{n}\frac{z_j}{z_k}\right)=0$ (d) $\text{Im}\left(\sum_{j=1}^{n}\sum_{k=1}^{n}\frac{z_j}{z_k}\right)=1$

11. $f:[0,1]\to R$ is a differentiable function such that $f(0)=0$ and $|f'(x)|\le k|f(x)|$ for all $x \in [0,1]$, $(k>0)$, then which of the following is/are always true?

(a) $f(x)=0,\ \forall x\in R$ (b) $f(x)=0,\ \forall x\in[0,1]$

(c) $f(x)\neq 0,\ \forall x\in[0,1]$ (d) $f(1)=k$

12. If $\log_2\left(\log_{\frac{1}{2}}(\log_2(x))\right)=\log_3\left(\log_{\frac{1}{3}}(\log_3(y))\right)=\log_5\left(\log_{\frac{1}{5}}(\log_5(z))\right)=0$ for positive x, y and z, then which of the following is/ are NOT true?

(a) $z<x<y$ (b) $x<y<z$

(c) $y<z<x$ (d) $z<y<x$

SECTION–III
Comprehension Type

This section contains 1 group of questions. Each group has 3 multiple choice questions based on a paragraph. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

Consider a convex quadrilateral *ABCD* with diagonals intersecting at *P*. Ellipse E_1 has foci *A* and *B* and is passing through *P*, ellipse E_2 has foci *C* and *D* and is also passing through *P*. Both E_1 and E_2 touch each other at *P*. This can be proved by contradiction assuming that E_1 and E_2 intersect at *P* and *Q* therefore by property of ellipse $PA+PB=QA+QB$, $PC+PD=QC+QD$. But if *Q* is not the point of intersection of diagonals, then $QA+QC>AC$ (triangle inequality) therefore *P* and *Q* are same points.

13. Ellipse E_3 has foci *B* and *C* and passes through *P*, then which of the following is incorrect?

(a) E_3 is orthogonal to E_1 at *P* (b) E_3 is orthogonal to E_2 at *P*

(c) circumcircle of BCP touches E_3 at P (d) E_3 has eccentricity $\frac{BC}{(BP+PC)}$

14. If $ABCD$ is a cyclic quadrilateral and eccentricities of E_1, E_2 and E_3 are e_1, e_2 and e_3 respectively, then

(a) $e_1 = e_2 = e_3$ (b) $e_1 = e_2$

(c) $e_3 = e_1$ (d) $e_3 = e_2$

15. If the angle bisector of $\angle BPA$ intersects AB and CD at M and N respectively, then $\frac{(MB+CN)}{BC}$ is equal to

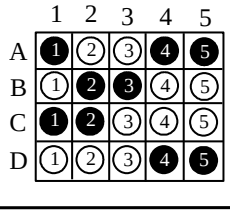
(a) $\frac{e_1}{e_3}$ (b) $\frac{e_3}{e_1}$

(c) $e_1 e_3$ (d) $\frac{1}{e_1 e_3}$

SECTION–IV
Matrix–Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled 1, 2, 3, 4 and 5. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – 1, 4 and 5; B – 2 and 3; C – 1 and 2; and D – 4 and 5; then the correct darkening of bubbles will look like the following.



1.

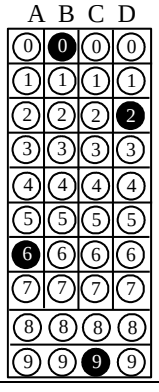
Column I	Column II
(A) $f(x)$ is defined for $x \geq 0$ and has a continuous derivative such that $f(0)=1;\ f'(0)=0$ and $(1+f(x))f''(x)=1+x$, then $f(x)$ is a/an	**1.** cubic polynomial function
(B) If $f(x).f(y)=f(x+y),\ \forall x,\ y \in R$ and $f(0)\neq 0,\ f(1)=e,\ f'(0)=1$, then $f(x)$ is a/an	**2.** increasing function
(C) If $f(x)$ is polynomial function satisfying $f(0)=0$ and $f(x^2+1)=f^2(x)+1$, then $f(x)$ is a/an	**3.** periodic function
(D) If f is a polynomial function such that $f: R^+ \to R^+$ and $f(f(x))=6x-f(x)$, then $f(x)$ is a/an	**4.** linear polynomial function **5.** quadratic polynomial function

2. Triangle is formed by joining the vertices of a cube.

	Column I	Column II
(A)	Probability that the triangle is isosceles (including equilateral) is	**1.** $\frac{3}{7}$
(B)	Probability that the triangle is isosceles (excluding equilateral) is	**2.** $\frac{4}{7}$
(C)	Probability that the triangle is scalene is	**3.** $\frac{6}{7}$
(D)	Probability that the triangle is right angled triangle is	**4.** $\frac{3}{14}$ **5.** $\frac{5}{14}$

SECTION–V
Integer Answer Type

This section contains 4 questions. The answer to each of the questions is a **single digit integer,** ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers A, B, C and D (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the following:



1. Let $p(x) = x^5 + x^2 + 1$ have roots x_1, x_2, x_3, x_4 and x_5 , $g(x) = x^2 - 2$, then find the value of $g(x_1)g(x_2)g(x_3)g(x_4)g(x_5) - 30g(x_1x_2x_3x_4x_5)$.

Space for work

mathongo

2. If $\alpha = e^{i2\pi/7}$ and $f(x) = A_0 + \sum_{k=1}^{20} A_k x^k$ and the value of

$f(x) + f(\alpha x) + f(\alpha^2 x) + ... + f(\alpha^6 x)$ is $k(A_0 + A_7 x^7 + A_{14} x^{14})$, then find the value of k.

Space for work

mathongo

3. $f : R \to R$ is $f(x) = \ln(x + \sqrt{x^2 + 1})$, then find the number of solutions to the equation $\left| f^{-1}(x) \right| = e^{-|x|}$.

Space for work

mathongo

4. $A(0, 2)$, B and C are points on parabola $y^2 = x + 4$ and such that $\angle CBA = \frac{\pi}{2}$, then find the least positive value of ordinate of C.

Space for work

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